

Specification
for
Small Horizontal CNC Lathe

Arnold Air Force Complex, AEDC

The specific machine needed is the Haas ST20

Provide all materials, labor, and equipment necessary to provide a new fully functional Horizontal CNC Lathe. Refurbished or remanufactured parts are not authorized for this requirement.

Small Horizontal CNC Lathe Machine specifications:

- Machine construction
 - Base, Bed and Headstock shall be Meehanite Cast Iron.
 - Slant Bed style to aid in coolant and chip fall.
 - Protective covers on all guideways.
 - Internal chip conveyor and coolant tank.
- Main Spindle and Headstock Specification
 - 20 Hp rating.
 - Variable speed to 4000 RPM Range with adequate high-speed bearings.
 - Spindle capacity shall be at least 3.5", bar capability to 2.5" diameter min.
 - Standard 3-jaw chuck size 10" min.
 - Hydraulic chuck.
 - A2-6 Spindle nose mount.
- Turret
 - 12 station 24 position indexing.
- Controls
 - Manufacturers latest version with 10" minimum Color LCD.
 - Controls cabinet shall be cooled.
 - Minimum 1GB program memory.
 - DNC Capability.
 - Programming should be compatible with FANUC protocol.
 - Controls equipped with conversational programming capability.
 - Helical interpolation and Custom Macro B.
 - AC Digital Servo & Spindle Drivers.
 - High Speed Machining Capability.
 - Data Server 256MB (DNC) min.

Not Sensitive

- Machine shall be equipped with Renishaw style probe tooling for part inspection.
- Remove / Disconnect Wi-Fi.
- Ethernet Connectivity.
- Manual pulse generator (Hand wheel).
- Spindle, feed and rapid override.
- Graph run simulator.
- Early power failure detection.
- Rigid tapping.
- Power surge protection.
- Travels
 - X axis at least 8"
 - Z axis at least 22"
 - Repeatability for X & Z Axis shall be +/- .0001"
 - Positioning for X & Z Axis shall be +/- .0001"/12"
- General
 - Max Swing at least 20".
 - Turning diameter to be 11" minimum.
 - Turning length to be 20" minimum.
 - Auto Lubrication system.
 - Internal coolant tank.
 - Full coverage removable splash guards.
 - Automatic Tool Setter, manual and automatic modes.
 - 1 each 1" face groover holder.
 - 2 each 1" ID holder.
 - 3 each 1.5" ID holder.
 - 1 each coolant block.
 - 9 each turret face wedge/clamp.
 - Chip conveyor.
 - Fully programmable tailstock.
 - A2-6 QG-80 Accu- Length Collet Chuck.
- Contractor shall provide:
 - Machine with lifting, leveling, and floor attachment hardware.
 - Machine built to NFPA 79 and operate with AEDC power system which is 480V, 3PH, 60 HZ or quote package shall include step down transformer to machine operation voltage.
 - Operation Manual, Maintenance Manual, Electrical Manual, & Schematics.
 - Standard Bar feed unit and parts catcher shall be included.
 - Standard tooling package turret tool holders applicable for this machine. To include Tooling Voucher.
 - Hydraulic Collet Chuck with Collet Package.
 - Integral coolant tank and pump.
 - Post Processor for Esprit CAM software.

Not Sensitive

- Verification
 - After delivery the Contractor shall test run the machine at the site in accordance with functional tests for accuracy required by ISO/ANSI Standards.
- Warranty
 - This machine shall be warranted against any defects in material, parts, and workmanship for 24 months after the date of installation. The control, servomotors and main spindle motor are warranted against defects in material, parts, and workmanship for 24 months after the date of installation.
- Maintenance, Parts and Service
 - The Contractor or representative shall be capable of responding to a repair service call on-site within 36 hours and repair parts delivery within 48 hours after placement of order.
- Packaging/Delivery
 - The Contractor shall provide all preservation, packaging, and packing to ensure safe delivery of the machine and related components to AEDC (FOB Arnold Air Force Base, TN). Deliver Machine to AEDC for off-loading by AEDC personnel.
- Installation/Training
 - Installation will be at the Arnold Air Force Base, TN; Arnold Engineering Development Complex's Model Shop, West Shop, Building 451. Installation of the machine shall be accomplished by Model Shop personnel and supervised by factory trained servicemen. This service shall be included in machine cost. Installation kit shall be provided in machine cost. Two-day initial training on machine features and operation shall be accomplished at machine startup and shall be included in machine cost.